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Python Sets

A Collection of Unique Elements

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What is a Set?

- A **set** is an **unordered** collection of **unique elements** in Python.
- Elements in a set do not allow duplicates.
- Sets are **mutable** (can be modified).
- Sets are written using curly braces { } or the **set()** function.
- Elements can be of different data types.

Creating a Set

```
# Using curly braces
s1 = {1, 2, 3, 4}
print(s1)

# Using set() function
s2 = set([10, 20, 30])
print(s2)

# Mixed data types
s3 = {1, "hello", 3.5}
print(s3)
```

Output:

```
{1, 2, 3, 4}
{10, 20, 30}
{1, 3.5, 'hello'}
```

Important Points

- Elements are unique (duplicates are removed).
- Sets are unordered (no indexing).
- Sets are mutable (can add or remove elements).
- Sets cannot be accessed using index (s[0] will give an error).
- Allows different data types.
- Useful for mathematical operations like union, intersection, difference, etc.

Set Operations

Operation	Symbol / Method	Example	Result
Union	| or union()	{1, 2} | {2, 3}	{1, 2, 3}
Intersection	& or intersection()	{1, 2} & {2, 3}	{2}
Difference	- or difference()	{1, 2} - {2, 3}	{1}
Symmetric Difference	^ or symmetric_difference()	{1, 2} ^ {2, 3}	{1, 3}

Adding and Removing Elements

```
s = {1, 2, 3}
s.add(4)       # Add element
print(s)
s.remove(2)    # Remove element (error if not present)
print(s)
s.discard(10)  # Removes element (no error if not present)
s.clear()      # Remove all elements
print(s)
```

Practice Question

What will be the output of the following code?

```
s = {10, 20, 20, 30, 40}
s.add(50)
s.remove(30)
print(s)
```

A {10, 20, 30, 40}

B {10, 20, 40, 50}

C {20, 30, 40, 50}

D {10, 20, 30, 40, 50}

Answer: B

Explanation:

Initially, duplicates are removed:

```
s = {10, 20, 30, 40}
```

After `s.add(50)`:

```
s = {10, 20, 30, 40, 50}
```

After `s.remove(30)`:

```
s = {10, 20, 40, 50}
```

So, the output is {10, 20, 40, 50}.

Key Takeaways

- Sets store only unique elements.
- Sets are unordered.
- No indexing (s[0] gives error).
- Use add(), remove(), discard(), clear() to modify sets.
- Support mathematical operations (union, intersection, difference, symmetric difference).
- Useful in real-world problems like finding common items, removing duplicates, etc.

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By Satyarthi

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Python Set Operations

Small Steps Big Skills!

Perform Operations on Sets

1. Union (| or union())

Returns all unique elements from both sets.

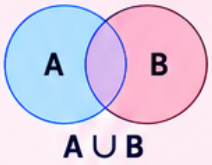
A | B or A.union(B)

Example Code:

```
A = {1, 2, 3}
B = {3, 4, 5}
print(A | B)   # or A.union(B)
```

Output:

```
{1, 2, 3, 4, 5}
```



2. Intersection (& or intersection())

Returns elements that are common in both sets.

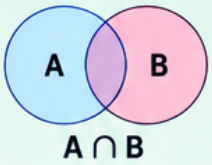
A & B or A.intersection(B)

Example Code:

```
A = {1, 2, 3}
B = {3, 4, 5}
print(A & B)  # or A.intersection(B)
```

Output:

```
{3}
```



3. Difference (- or difference())

Returns elements in **A** that are not in **B**.

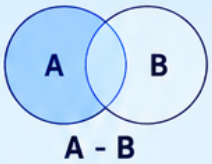
A - B or A.difference(B)

Example Code:

```
A = {1, 2, 3}
B = {3, 4, 5}
print(A - B)  # or A.difference(B)
```

Output:

```
{1, 2}
```



4. Reverse Difference

Returns elements in **B** that are not in **A**.

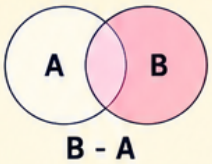
B - A or B.difference(A)

Example Code:

```
A = {1, 2, 3}
B = {3, 4, 5}
print(B - A)  # or B.difference(A)
```

Output:

```
{4, 5}
```



5. Symmetric Difference (^ or symmetric_difference())

Returns elements that are in either set, but not in both.

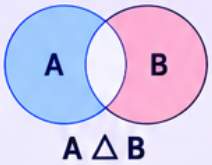
A ^ B or A.symmetric_difference(B)

Example Code:

```
A = {1, 2, 3}
B = {3, 4, 5}
print(A ^ B)  # or A.symmetric_difference(B)
```

Output:

```
{1, 2, 4, 5}
```



6. Adding an Element (add())

Adds a single element to the set.

S.add(x)

Example Code:

```
S = {1, 2, 3}
S.add(4)
print(S)
```

Output:

```
{1, 2, 3, 4}
```

Adds only one element. Duplicates are ignored.

7. Removing an Element (remove())

Removes a specific element (fom the set (raises an error if not present).

S.remove(x)

Example Code:

```
S = {1, 2, 3}
S.remove(2)
print(S)
```

Output:

```
{1, 3}
```

If the element is not present, it raises a **KeyError**.

8. Discard an Element (discard())

Removes a specific element (fom the set (no error if not present).

S.discard(x)

Example Code:

```
S = {1, 2, 3}
S.discard(5)
print(S)
```

Output:

```
{1, 2, 3}
```

If the element is not present, no error occurs.

9. Clear Set (clear())

Removes all elements from the set.

S.clear()

Example Code:

```
S = {1, 2, 3}
S.clear()
print(S)
```

Output:

```
set()
```

Makes the set empty.

10. Length of a Set (len())

Returns the number of elements in the set.

len(S)

Example Code:

```
S = {10, 20, 30, 40}
print(len(S))
```

Output:

```
4
```

Counts only unique elements.

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By Satyarthi

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Python Lists & Tuples

Practice Question

Topic: Lists vs Tuples

Small Steps Big Skills!

Question

What will be the output of the following code?

```
my_list = [1, 2, 3]
my_tuple = (1, 2, 3)
my_list[0] = 100
print(my_list)
print(my_tuple)
```

A
```
[100, 2, 3]
(1, 2, 3)
```

B
```
[1, 2, 3]
(100, 2, 3)
```

C
```
[100, 2, 3]
(100, 2, 3)
```

D
Error: 'tuple' object does not support item assignment

Lists are mutable, Tuples are immutable!

Answer: A

Explanation:

- Lists are mutable, so the first element can be changed.
- Tuples are immutable, so the value remains unchanged.
- The output will be:

```
[100, 2, 3]
(1, 2, 3)
```

Visual Representation

List (Mutable)

my_list = [1, 2, 3]

0	1	2
1	2	3

my_list[0] = 100

0	1	2
100	2	3

Tuple (Immutable)

my_tuple = (1, 2, 3)

0	1	2
1	2	3

✗ Cannot change value (Immutable)

Key Points

- Lists are mutable (can be changed).
- Tuples are immutable (cannot be changed).
- Both can store similar data types.
- Use lists when you need to modify data.
- Use tuples when you want data to remain constant (e.g., coordinates, fixed records).

Remember

- List → [] (square brackets)
- Tuple → () (parentheses)
- List methods: append(), remove(), etc.
- Tuple methods: count(), index(), etc.
- Try changing values in both and see the difference!

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Python Sets

Practice Question

Test Your Understanding

Small Steps Big Skills!

Question

What will be the output of the following code?

```
s = {1, 2, 3, 4}
s.add(4)
s.add(5)
s.remove(2)
print(s)
```

A {1, 3, 4, 5}

B {1, 2, 3, 4, 5}

C {1, 2, 4, 5}

D {3, 4, 5}

Think Logically Apply Concepts Choose the Correct Output!

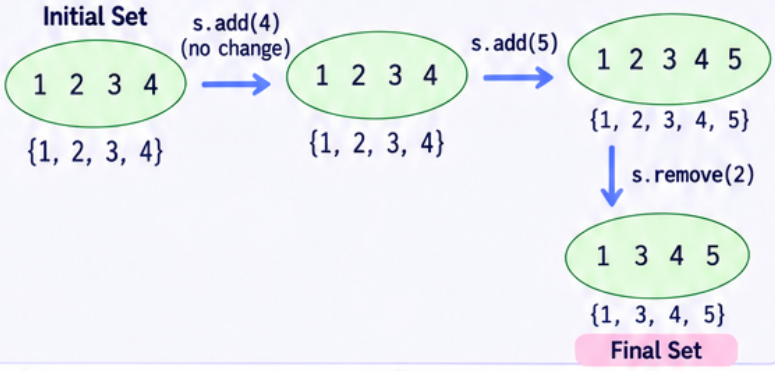
Answer: A

Explanation:

- **Initial set:**
  `s = {1, 2, 3, 4}`
- **s.add(4):**
  4 is already present, so no change
  `s = {1, 2, 3, 4}`
- **s.add(5):**
  `s = {1, 2, 3, 4, 5}`
- **s.remove(2):**
  `s = {1, 3, 4, 5}`
- Finally, `print(s)` will output:

```
{1, 3, 4, 5}
```

Visual Representation



Key Points

- A set is an unordered collection of unique elements.
- Duplicate elements are ignored.
- Use add() to insert an element.
- Use remove() to delete an element (raises error if not present).
- Sets are mutable.
- The order of elements in the output may vary.

Related Functions

- add() – Add an element
- remove() – Remove an element (error if not present)
- discard() – Remove an element (no error if not present)
- union() – Combine two sets
- intersection() – Common elements
- difference() – Elements in A not in B
- clear() – Remove all elements
- len() – Number of elements

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Python Sets

Practice Question

Topic: Set Operations

Small Steps Big Skills!

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Question

What will be the output of the following code?

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
C = A | B       # union of A and B
D = A & B       # intersection of A and B
E = A - B       # elements in A but not in B
F = B - A       # elements in B but not in A
print(C, D, E, F)
```

Apply Set Concepts and Find the Output!

A `{1, 2, 3, 4, 5, 6} {3, 4} {1, 2} {5, 6}`

B `{1, 2, 3, 4} {3, 4} {1, 2} {5, 6}`

C `{1, 2, 3, 4, 5, 6} {1, 2} {3, 4} {5, 6}`

D `{3, 4} {1, 2, 5, 6} {1, 2} {5, 6}`

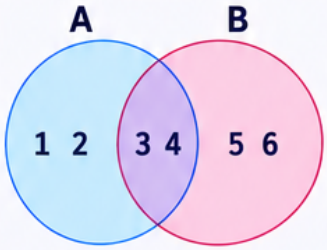
Answer: A

Explanation:

- **A | B** gives the union of both sets (all unique elements).
  `C = {1, 2, 3, 4, 5, 6}`
- **A & B** gives the common elements (intersection).
  `D = {3, 4}`
- **A - B** gives elements in A but not in B.
  `E = {1, 2}`
- **B - A** gives elements in B but not in A.
  `F = {5, 6}`
- **So, the output will be:**

```
{1, 2, 3, 4, 5, 6} {3, 4} {1, 2} {5, 6}
```

Set Visualization



A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
A | B = {1, 2, 3, 4, 5, 6}
A & B = {3, 4}
A - B = {1, 2}
B - A = {5, 6}

Key Takeaways

- Sets store unique elements (no duplicates).
- Common set operations: union (|), intersection (&), difference (-).
- Sets are unordered, so output order may vary.
- Use sets when you need to work with unique data and perform mathematical operations.
- Very useful in real-world problems like finding common items, removing duplicates, etc.

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Python Sets

Practice Question

Test Your Knowledge

Small Steps Big Skills!

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Question

What will be the output of the following code?

```
A = {1, 2, 3, 4}
B = {3, 4, 4, 5}
A.update(B)
B.add(6)
A.remove(2)
print(A)
print(B)
```

Understand Sets Solve Better!

A

```
{1, 2, 3, 4, 5}
{3, 4, 5, 6}
```

B

```
{1, 3, 4, 5}
{3, 4, 5, 6}
```

C

```
{1, 3, 4, 5, 6}
{3, 4, 6}
```

D

```
{1, 3, 4, 6}
{3, 4, 5, 6}
```

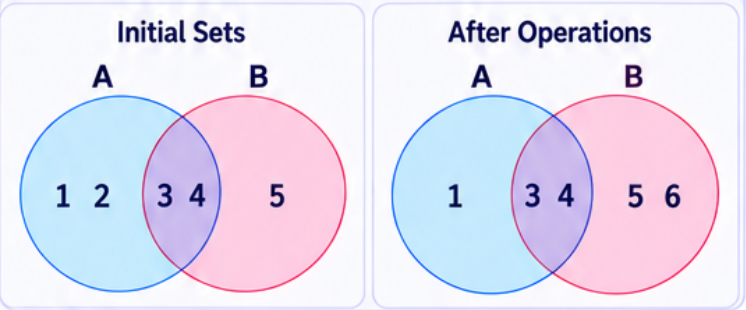
Answer: B

Explanation:

- **Initially:**
  `A = {1, 2, 3, 4}`
  `B = {3, 4, 5}` (duplicate 4 is ignored)
- **A.update(B):**
  A becomes `{1, 2, 3, 4, 5}`
  (adds all elements from B)
- **B.add(6):**
  B becomes `{3, 4, 5, 6}`
- **A.remove(2):**
  2 is removed from A
  A becomes `{1, 3, 4, 5}`
- **Final output:**

```
{1, 3, 4, 5}
{3, 4, 5, 6}
```

Step-by-Step Visualization



Key Takeaways

- Sets store unique elements (no duplicates).
- update() adds all elements from another set.
- add() adds a single element.
- remove() removes an element (raises error if not present).
- Set elements are unordered (output order may vary).

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Learn Practice Improve Grow

Python Dictionaries

Store Data in Key-Value Pairs

A Powerful & Flexible Data Structure

Small Steps Big Skills!

What is a Dictionary?

- A dictionary in Python is a collection of key-value pairs.
- It is unordered, mutable and allows unique keys.
- Keys can be strings, numbers or tuples (immutable).
- Values can be of any data type.

Syntax:

```
dict_name = {key1: value1,
             key2: value2,
             ...}
```

Creating a Dictionary

```
>>> student = {"name": "Aman",
               "age": 19,
               "course": "BPT"}
>>> print(student)
{'name': 'Aman', 'age': 19,
'course': 'BPT'}
```

Accessing Values

- Use keys to access values.

```
>>> student = {"name": "Aman",
               "age": 19}
>>> print(student["name"])
Aman
>>> print(student["age"])
19
```

Adding or Updating Items

- Use **key = value** to add a new item or update an existing one.

```
>>> student = {"name": "Aman"}
>>> student["age"] = 19   # add
>>> student["name"] = "Sundram"
    # update
>>> print(student)
{'name': 'Sundram', 'age': 19}
```

Removing Items

- **pop()** – removes item by key
- **del** – removes item by key
- **clear()** – removes all items

```
>>> student = {"name": "Aman",
               "age": 19}
>>> student.pop("age")
19
>>> print(student)
{'name': 'Aman'}
>>> del student["name"]
>>> print(student)
{}
```

Common Dictionary Methods

Method	Description
get(key)	Returns value (no error if key not found)
keys()	Returns all keys
values()	Returns all values
items()	Returns all key-value pairs
update()	Updates with another dictionary
pop(key)	Removes and returns value for the key
clear()	Removes all items
len()	Returns number of items

Looping Through a Dictionary

- You can iterate using keys, values or items.

```
>>> student = {"name": "Aman",
               "age": 19,
               "course": "BPT"}
>>> for key in student:       # keys
...     print(key, ":", student[key])
name : Aman
age : 19
course : BPT
```

Nested Dictionary

- A dictionary can contain another dictionary.

```
>>> students = {
...     "s1": {"name": "Aman", "age": 19},
...     "s2": {"name": "Riya", "age": 20}
... }
>>> print(students["s1"]["name"])
Aman
```

Example: Student Marks

```
>>> marks = {"Physics": 85,
             "Chemistry": 78,
             "Biology": 92}
>>> print(marks["Biology"])
92
>>> marks["Maths"] = 88
>>> print(marks)
{'Physics': 85, 'Chemistry': 78,
'Biology': 92, 'Maths': 88}
```

Practice Question

Create a dictionary with the following details and perform the given tasks.

```
student = {"name": "Rohit", "age": 18, "course": "BSc"}
```

1. Add a new key "city" with value "Patna".
2. Update the age to 19.
3. Remove the "course" key.
4. Print the final dictionary.

Expected Output

```
{'name': 'Rohit',
'age': 19,
'city': 'Patna'}
```

Key Takeaways

- Dictionary = key-value pairs.
- Keys are unique and immutable.
- Values can be any data type.
- Dictionaries are mutable.
- Useful for real-world data (e.g., student records, contact lists, configuration).

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Python Dictionaries

Two Important Concepts

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1 Creating a Dictionary

- A dictionary is a collection of key-value pairs written inside curly braces { }.
- Each item is in the form key : value, separated by commas.
- Keys must be unique and can be of any immutable type (string, number, tuple).
- Values can be of any data type.

Syntax:

```
dictionary_name = {key1: value1,
                   key2: value2,
                   ...}
```

Example:

```
# Creating a dictionary
student = {
    "name": "Aman",
    "age": 19,
    "course": "BPT"
}
print(student)
```

Output:

```
{'name': 'Aman', 'age': 19,
'course': 'BPT'}
```

2 Using dict() Constructor

- You can also create a dictionary using the built-in dict() function.
- Pass key=value pairs inside the parentheses, separated by commas.
- It is an alternative way to create a dictionary and is useful when creating a dictionary programmatically.

Syntax:

```
dictionary_name = dict(key1=value1,
                       key2=value2,
                       ...}
```

Example:

```
# Creating a dictionary using dict()
student = dict(name="Rohit",
               age=18,
               course="BSc")

print(student)
```

Output:

```
{'name': 'Rohit', 'age': 18,
'course': 'BSc'}
```

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Python Dictionaries

Next 2 Important Concepts

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3 Accessing Dictionary Items

You can access values in a dictionary using the key. If the key is not present, it will raise a **KeyError**.

Syntax:

```
dictionary_name[key]
# or
dictionary_name.get(key)  # safer method
```

Example:

```
# Creating a dictionary
student = {
    "name": "Aman",
    "age": 19,
    "course": "BPT"
}
# Accessing values
print(student["name"])      # Aman
print(student["age"])       # 19
print(student.get("course"))  # BPT

# Trying to access a key that doesn't exist
print(student.get("city", "Not Found"))
# Not Found
```

Output:

```
Aman
19
BPT
Not Found
```

4 Updating and Adding Items

You can change the value of an existing key (update), or add a new key-value pair to a dictionary.

Syntax:

```
dictionary_name[key] = value  # update
dictionary_name[new_key] = value  # add
```

Example:

```
# Creating a dictionary
student = {
    "name": "Aman",
    "age": 19,
    "course": "BPT"
}

# Updating an existing value
student["age"] = 20
# age updated from 19 to 20

# Adding a new key-value pair
student["city"] = "Patna"
# new key added

print(student)
```

Output:

```
{'name': 'Aman', 'age': 20,
'course': 'BPT', 'city': 'Patna'}
```

Key Takeaways

- Use dictionary_name[key] to access a value.
- Use get(key, default) to avoid errors.
- You can update the value of an existing key.
- You can add new key-value pairs easily.
- Keys must be unique.
- Dictionaries are mutable (can be changed).

Practice Build Concepts Do Better!

By Satyarthi

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Python Dictionaries

Next 2 Important Concepts

Small Steps Big Skills!

5 Dictionary Methods

Python provides several built-in methods to work with dictionaries, such as `keys()`, `values()`, `items()`, `get()`, etc.

Method	Description
`keys()`	Returns all the keys
`values()`	Returns all the values
`items()`	Returns all key-value pairs (as tuples)
`get(key)`	Returns value for the key (if key not found, returns None or a default value)
`update()`	Updates the dictionary with another dictionary
`pop(key)`	Removes the specified key and returns its value
`clear()`	Removes all items
`len()`	Returns the number of items

Example:

```
student = {"name": "Aman",
           "age": 19,
           "course": "BPT"}
print(student.keys())
print(student.values())
print(student.items())
print(student.get("age"))
```

Output:

```
dict_keys(['name', 'age', 'course'])
dict_values(['Aman', 19, 'BPT'])
dict_items([('name', 'Aman'), ('age', 19),
('course', 'BPT')])
19
```

6 Looping Through a Dictionary

You can iterate through a dictionary using a for loop to access keys, values, or both keys and values.

1. Looping through keys

```
student = {"name": "Aman",
           "age": 19,
           "course": "BPT"}

for key in student:
    print(key)
```

Output:

```
name
age
course
```

2. Looping through values

```
for value in student.values():
    print(value)
```

Output:

```
Aman
19
BPT
```

3. Looping through key-value pairs

```
for key, value in student.items():
    print(key, ":", value)
```

Output:

```
name : Aman
age : 19
course : BPT
```

4. Using items() in a different way

```
for item in student.items():
    print(item)
```

Output:

```
('name', 'Aman')
('age', 19)
('course', 'BPT')
```

Key Takeaways

- Use built-in methods to easily work with dictionaries.
- You can loop through keys, values or both.
- `items()` returns key-value pairs as tuples.
- Dictionaries are flexible and very useful for real-world data (e.g., student records, contact lists, configuration).
- Practice these concepts with examples to strengthen your understanding.

Practice Today, Build a Better Tomorrow!

By Satyarthi

Code Practice Grow

Python String Methods

Practice Question 3

Small Practice Big Improvement!

Question 3

Topic: **replace()** method

What will be the output of the following code?

```
s = "I love Python"
print(s.replace("Python", "Java"))
```

A 'I love Java'

B 'I Love Java'

C 'I love python'

D 'I love Python Java'

"Same Method, New Possibilities!"

Answer: A

Explanation:

The **replace()** method returns a new string in which all occurrences of the specified substring are replaced with another substring.

Here, **"Python"** is replaced by **"Java"**, so the output is **'I love Java'**.

The original string remains unchanged.

Key Takeaway

- **replace()** replaces all occurrences of a substring.
- It is case-sensitive.
- It does not modify the original string (strings are immutable in Python).

```
s = "apple apple"
print(s.replace("apple", "orange"))
# Output: orange orange
```

Practice Today, Stronger Tomorrow!

By Satyarthi

Learn Practice Improve Grow

Python Dictionaries

Next 2 Important Concepts

Small Steps Big Skills!

Keep Practicing Python!

7 Removing Items from a Dictionary

You can remove an item from a dictionary using `pop()`, `popitem()`, `del`, or `clear()`.
These methods help you delete a specific key, the last item, or all items.

1. Using pop(key)

Removes the specified key and returns its value.
If the key is not found, it raises a `KeyError`.

```
student = {"name": "Aman",
"age": 19, "course": "BPT"}
removed = student.pop("age")
print(removed)
print(student)
```

Output:

```
19
{'name': 'Aman',
'course': 'BPT'}
```

2. Using popitem()

Removes and returns the last inserted key-value pair (as a tuple).

```
student = {"name": "Aman",
"age": 19}
item = student.popitem()
print(item)
print(student)
```

Output:

```
('age', 19)
{'name': 'Aman'}
```

3. Using del

Removes the specified key (does not return the value).

```
student = {"name": "Aman",
           "age": 19}
del student["name"]
print(student)
```

Output:

```
{'age': 19}
```

4. Using clear()

Removes all items from the dictionary.

```
student = {"name": "Aman",
           "age": 19}
student.clear()
print(student)
```

Output:

```
{}
```

8 Dictionary Comprehension

You can create a dictionary using dictionary comprehension, similar to list comprehension.
It provides a short and clean way to create dictionaries based on a condition or expression.

1. Creating a dictionary with squares (1 to 5)

```
squares = {x: x**2 for x in range(1, 6)}
print(squares)
```

Output:

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

2. Creating a dictionary with a condition

```
even_squares = {x: x**2 for x in range(1, 11)
                if x % 2 == 0}
print(even_squares)
```

Output:

```
{2: 4, 4: 16, 6: 36, 8: 64, 10: 100}
```

3. Creating a dictionary from two lists

```
keys = ["name", "age", "city"]
values = ["Aman", 19, "Patna"]
student = {keys[i]: values[i] for i in range(len(keys))}
print(student)
```

Output:

```
{'name': 'Aman', 'age': 19, 'city': 'Patna'}
```

Key Takeaways

- Use pop(), popitem(), del, or clear() to remove items.
- Dictionary comprehension helps you create dictionaries in a single line.
- You can use conditions inside dictionary comprehension.
- Dictionaries are flexible and useful for real-world data handling.
- Practice with different examples to strengthen your concepts.

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Python Dictionaries

Practice Questions

Small Steps Big Skills!

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1 Question 1

Consider the following dictionary:

```
student = {
    "name": "Aman",
    "age": 19,
    "course": "BPT"
}
```

What will be the output of the following code?

```
print(student.get("city", "Not Found"))
```

A. None

B. Not Found

C. Error

D. "city"

Concept Revision:

- The `get()` method returns the value for a key if it exists.
- If the key is not found, it returns the specified default value (here `"Not Found"`).
- It does not raise an error.

2 Question 2

What will be the output of the following code?

```
marks = {"Physics": 85, "Chemistry": 78}
marks["Biology"] = 92
marks["Chemistry"] = 80
del marks["Physics"]
print(marks)
```

A. `{'Physics': 85, 'Chemistry': 78, 'Biology': 92}`

B. `{'Chemistry': 80, 'Biology': 92}`

C. `{'Physics': 85, 'Chemistry': 80, 'Biology': 92}`

D. `{'Chemistry': 78, 'Biology': 92}`

Concept Revision:

- You can add a new key-value pair using `dict[key] = value`.
- If a key already exists, its value gets updated.
- You can remove a key using `del dict[key]`.
- The final dictionary contains the remaining key-value pairs.

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